

Full stack robotics engineer with with a skill set ranging from mechanical design, to embedded software development. I am a student at Worcester Polytechnic Institute on an endless mission to understand the inner workings of everything.

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SKILLS

- Programming Languages: C / C++ , Zig, Python, Java, Kotlin, Nix
- Software Libraries: ImGui, Eigen, ROS, OpenCV, Numpy
- CAD: OnShape, KiCAD, FreeCAD, Solidworks
- Operating Systems: NixOS, Arch, FreeRTOS, Ubuntu, Windows
- Manufacturing: 3d printing, CNC Router, CNC Mill, Laser Cutter
- Software Tools: Git, SSH, Vim, Gnu Debugger, Docker
- Equipment: Oscilloscope, Multimeter, Logic Analyzer, LCR Meter
- Hand Tools: Basic Woodworking, Electrical, and Assembly Tools

EDUCATION

B.S. - ROBOTICS ENGINEERING

WORCESTER POLYTECHNIC INSTITUTE
Expected Graduation May 2027

B.S. - ELECTRICAL AND COMPUTER ENGINEERING

WORCESTER POLYTECHNIC INSTITUTE
Expected Graduation May 2027

EXPERIENCE

HYDROFLEX

WPI MEDICAL MANUFACTURING AND INNOVATION LAB

JULY 2025 - PRESENT
WORCESTER, MA

- Electrical engineering for a remote driven internal polishing machine
- Assist with data collection and processing to detect proper grinding orbit using audio feedback
- Built and implemented controlller interface for adjusting live feed and speed
- Designed and prototyped test unit to be sent out for initial manufacturing process testing.

EMBEDDED FIRMWARE DEVELOPMENT

AGZEN

JULY 2025 - AUGUST 2025
SOMMERVILLE, MA

- Collaboratively designed point to point, node based embedded communication network structure
- Implemented prototype network on an STM32 with FreeRTOS over full duplex RS485 links
- Assembled and did initial firmware bringup on prototype sensor control PCB

WPI GOAT FAST RACING

WORCESTER POLYTECHNIC INSTITUTE

AUGUST 2024 - PRESENT
WORCESTER, MA

- Worked on a team of 15 plus engineers to build a fully electric racecar for the Formula Hybrid competition
- Coordinated design and mounting of low voltage sensor systems and communication
- Retrofit 400V accumulator to be completely manufactured in house in under 7 weeks
 - ▶ Created manufacturing and assembly process
 - ▶ Assembled full electro-mechanical system
 - ▶ Created tools to assist with assembly (Spot Welding Jig, Press for copper Bus Bars, Current Testing Jig)
 - ▶ designed custom 120A fusable links from 0.127mm copper sheets and empirically measured fusing curves
- Designed custom PCB wheel speed sensor using analog quadrature hall effect sensors to measure rotation continuously
- Worked on design of safety systems and procedures for high voltages

USGS STUDENT CONTRACTOR

UNITED STATES GEOLOGICAL SURVEY

MAY 2024 - AUGUST 2024
ANN ARBOR, MI

- Contributed to ROSBag data unpacking library to simplify and streamline programmatic data manipulation.
- Standardized conventions for robot position and orientation.
- Implemented 3D Point Cloud data processing and cleaning scripts to colocate a series of point clouds from GPS data.
- Worked on research vessels in the Great Lakes to gather scientific data with under water autonomous vehicles and sonar scanning technology.
- Processed images using a low pass filter to remove shadows and improve automated fish counting algorithms.

PROJECTS

TELOMETER

- Designed and implemented a telemetry protocol and visualization dashboard
- Implemented data synchronization, processing, and logging
- Designed to send serialized data with only 2 bytes of overhead per packet
- Used in robotics courses to enable real time data feedback and control loop tuning.
- Implemented a live front end plotting and visualization dashboard
- Created duplicate zig and cpp APIs

SOUND CAMERA

- Designed distributed phased array microphone sensors
- Manufactured custom PCB with RS485 interface for initial prototype
- Designed 3d audio reconstruction and replay software